

Survival Analysis Report

A Live Quarto Demo

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```
library(survival)
library(ggplot2)
library(kableExtra)
```

1 Introduction

This document demonstrates how Quarto integrates **code**, **prose**, and **mathematics** in a single reproducible file.

The dataset contains 228 patients from the North Central Cancer Treatment Group lung cancer study. A total of 165 deaths were observed over the follow-up period.

We use the Cox proportional hazards model (Cox 1972):

$$h(t \mid \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \boldsymbol{\beta}) \quad (1)$$

2 Descriptive Statistics

Table 1 shows the baseline characteristics of the study population.

```
df <- lung |>
  dplyr::summarise(
    n          = dplyr::n(),
    age_mean   = round(mean(age, na.rm = TRUE), 1),
    age_sd     = round(sd(age, na.rm = TRUE), 1),
    pct_male   = round(mean(sex == 1, na.rm = TRUE) * 100, 1),
    pct_events = round(mean(status == 2, na.rm = TRUE) * 100, 1)
  )

data.frame(
  Characteristic = c("Sample size", "Age, mean (SD)",
                    "Male sex, %", "Deaths, %"),
  Value = c(df$n,
            paste0(df$age_mean, " (", df$age_sd, ")"),
            paste0(df$pct_male, "%"),
            paste0(df$pct_events, "%"))
) |>
  kable(booktabs = TRUE, col.names = c("Characteristic", "Value")) |>
  kable_styling(latex_options = "hold_position")
```

Table 1: Baseline characteristics of lung cancer patients.

Characteristic	Value
Sample size	228
Age, mean (SD)	62.4 (9.1)
Male sex, %	60.5%
Deaths, %	72.4%

3 Kaplan-Meier Estimate

Figure 1 displays the overall Kaplan-Meier survival curve.

```
km <- survfit(Surv(time, status == 2) ~ 1, data = lung)

plot(km,
```

```

xlab = "Days", ylab = "Survival probability",
conf.int = TRUE, col = "#0072B2", lwd = 2)
abline(h = 0.5, lty = 2, col = "grey50")

```

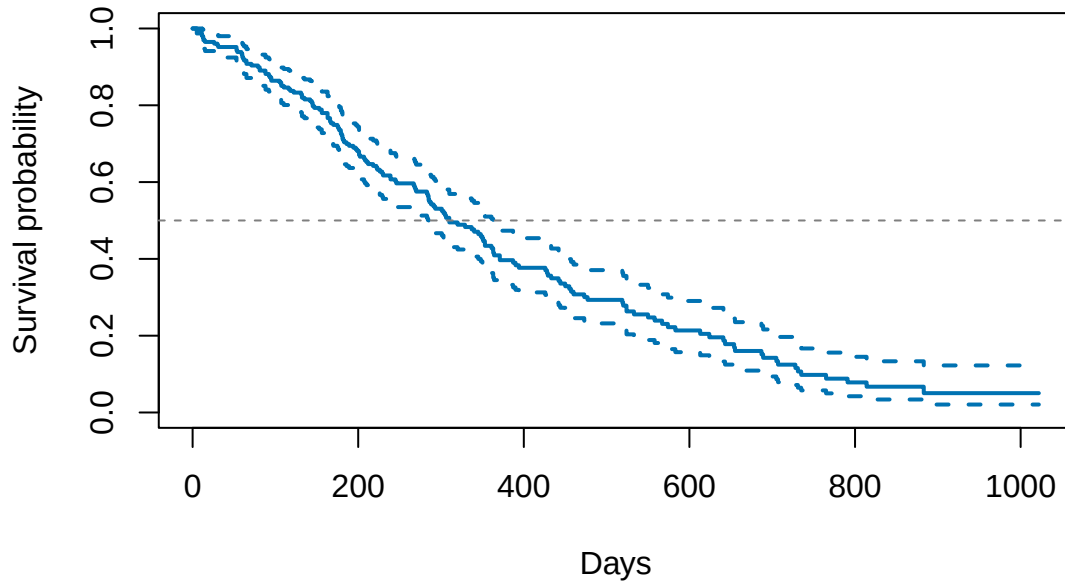


Figure 1: Kaplan-Meier estimate of overall survival.

The median survival was **310** days (95% CI: 285 – 363 days).

4 Cox Regression

```

fit <- coxph(Surv(time, status == 2) ~ age + sex + ph.ecog,
             data = lung)

broom::tidy(fit, exponentiate = TRUE, conf.int = TRUE) |>
  dplyr::mutate(dplyr::across(where(is.numeric), \(x) round(x, 3))) |>
  kable(
    booktabs = TRUE,
    col.names = c("Term", "HR", "Std. Error", "z", "p-value",
                  "95% CI Low", "95% CI High")
  ) |>
  kable_styling(latex_options = c("hold_position", "scale_down"))

```

Table 2: Cox proportional hazards model results.

Term	HR	Std. Error	z	p-value	95% CI Low	95% CI High
age	1.011	0.009	1.194	0.232	0.993	1.030
sex	0.575	0.168	-3.294	0.001	0.414	0.799
ph.ecog	1.590	0.114	4.083	0.000	1.273	1.986

5 Conclusion

Quarto renders this document to PDF and HTML from the same source. The analysis is fully reproducible: change the data → re-render → all numbers, figures, and tables update automatically.

References

Cox, D. R. 1972. “Regression Models and Life-Tables.” *Journal of the Royal Statistical Society: Series B* 34 (2): 187–220. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>.